HABIBA RAFIQUE

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SUMMARY

Undergraduate computer science student at BUET with growing interests in machine learning, digital health, and human-centered AI. Experienced in full-stack development. Currently engaging in academic research on self-tracking behavior and exploring preference-based reinforcement learning .

EDUCATION

Bangladesh University of Engineering and Technology (BUET) *Jan 2022 - Jun 2026*
BSc in Computer Science and Engineering

RESEARCH EXPERIENCE

Research Trainee (AI Research Bootcamp) *July 2025 - Dec 2025*
Advanced Machine Intelligence Research Lab (AMIRL) · Remote

- Conducted a systematic literature review (SLR), methodology, research synthesis for large language models, focusing on RLHF (Reinforcement Learning from Human Feedback) and DPO (Direct Preference Optimization) methods
- Analyzed online vs. offline alignment approaches, evaluating trade-offs in stability, computational efficiency, and generalization performance

A Comparative Study of Classical and Hybrid Quantum Machine Learning Models for Skin Lesion Classification using Latent Feature Embeddings
2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)

- Contributed to methodology design and experimental framework development
- Assisted with dataset preparation, MobileCLIP-based feature extraction, PCA-based dimensionality reduction, and hybrid quantum-classical model training
- Supported performance evaluation, including ROC analysis, confusion matrices, comparative performance tables, and manuscript drafting.

Undergraduate Thesis—Self-Tracking and Digital Health Among Bangladeshi Adults *2025–Present*

- Investigating behavioral patterns, self-tracking practices, and digital health adoption among Bangladeshi adults
- Applying quantitative and qualitative research methodologies for data analysis and interpretation

Research Internship Program — CHIRAL Bangladesh (Cohort 9.0)

- Developed foundational skills in literature review, research gap identification, scientific writing, and problem formulation

PROFESSIONAL AFFILIATIONS

General Member *Jan 2026–Present*
BRAC Human-Centered Computing and Society (HCCS)

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C/C++, Java, SQL
Frontend: React.js, HTML5, CSS3
Backend: Node.js, Express.js, MongoDB, PostgreSQL, Qdrant, REST APIs
ML/DL: Pandas, Numpy, Scikit-learn, Matplotlib, PyTorch, TensorFlow
Tools & Platforms: Git, Docker, Linux, VS Code, LaTeX, Figma
Soft Skills: Scientific writing, teamwork, analytical thinking, communication

EXPERIENCE

Learnathon 3.0: Fresher Training Program — Participant *Jan 2025 – Apr 2025*
Geeky Solutions · Apprenticeship (Remote)

- Completed intensive full-stack and problem-solving training through collaborative, industry-oriented projects.
- Strengthened foundational coding practices, version control workflows, and software design.

Achievements

- Secured the 1st Runner-Up position at the **Eco Tech Hackathon 2026**, organized by BUET.

PROJECTS

AI-Powered Learning Platform — StuddyBuddy GitHub
Technologies: Node.js, Express, PostgreSQL, Qdrant, React, Gemini API

- Developed an AI-assisted study platform generating quizzes and flashcards using Google Gemini.
- Built React frontend with Material-UI and PDF.js for interactive PDF viewing and annotation.
- Implemented semantic search with Qdrant and secure Google OAuth authentication.
- Deployed using Docker and Azure VM; automated CI/CD with GitHub Actions.

TourHub — Full-Stack Tour Booking Platform GitHub
Technologies: HTML, CSS, JavaScript, Node.js, Oracle DB, Bootstrap

- Built tour search, booking, payments, notifications, and user profile features.
- Designed admin panel for package management and participant tracking.

E-commerce Website GitHub
Technologies: MERN Stack

- Designed secure authentication and role-based admin dashboards.
- Built product catalog, cart, order system, and discount features.

ByteFixers — Project Management Tool GitHub
Technologies: MERN, Gemini AI API, Cloudinary

- Developed project management features: chat, JWT auth, subtasks, AI recommendations, activity logs.
- Integrated CI/CD (GitHub Actions) and unit testing (Jest).

Mission Antarctica Impossible GitHub
Technologies: C, C++, iGraphics

- Implemented optimized collision detection and interactive game behaviours.

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification – Coursera.
- Participant at Learnathon 3.0: Fresher Training Program by Geeky Solutions - Certificate
- Research Internship Program (Cohort 9.0), CHIRAL Bangladesh– CHIRAL.

LANGUAGES

- English: Fluent
- Bengali: Native